

Identification of ferroptosis biomarkers and immune infiltration landscapes in atrial fibrillation A bioinformatics analysis

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Abstract

Ferroptosis has been recognized as a critical factor in the development of atrial fibrillation (AF), but its precise mechanisms remain unclear. We downloaded the GSE115574 dataset from the gene expression omnibus database to analyze the expression levels of ferroptosis-related genes (FRGs) and identify differentially expressed genes (DEGs). Least absolute shrinkage and selection operator (LASSO) and support vector machine-recursive feature elimination (SVM-RFE) machine learning techniques were employed to identify key genes associated with AF. The diagnostic performance of these genes was evaluated using Receiver operating characteristic curves (ROC) and validated in an independent AF dataset. miRNA and lncRNA predictions for potential binding to these key genes were conducted using miRBase, miRDB, and TargetScan. Furthermore, gene set enrichment analysis (GSEA) enrichment analysis, immune cell infiltration analysis, and targeted drug prediction were performed. The intersection of LASSO regression and SVM-RFE analyses identified 7 DEGs significantly associated with AF. Validation through ROC and an additional dataset confirmed the importance of MAPK14, CAV1, and ADAM23. Significant infiltration of memory B cells, regulatory T cells, and monocytes was observed in atrial tissues. Seventy-two miRNAs were predicted to potentially target MAPK14, and 2 drugs were identified as targeting CAV1. This study underscores the involvement of FRGs in AF through machine learning and validation approaches. The observed immune cell infiltration suggests a potential link between immune response and AF. The predicted ceRNA network offers new insights into gene regulation, presenting potential biomarkers and therapeutic targets for AF.

Abbreviations: AF = atrial fibrillation, AUC = area under the ROC curve, DEGs = differentially expressed genes, FRGs = ferroptosis-related genes, GEO = gene expression omnibus, GO = gene ontology, GSEA = gene set enrichment analysis, KEGG = Kyoto Encyclopedia of Genes and Genomes, LASSO = least absolute shrinkage and selection operator, ROC = receiver operating characteristic curves, SVM-RFE = support vector machine-recursive feature elimination.

Keywords: atrial fibrillation, ferroptosis, lasso regression, machine learning, SVM-RFE

1. Introduction

Atrial fibrillation (AF) is 1 of the most common cardiac arrhythmias, affecting millions of people globally. It significantly elevates the risk of stroke, heart failure, and cardiovascular mortality, placing a considerable burden on both patients and healthcare systems.^[1] Despite the availability of treatment strategies such as anticoagulation therapy and rhythm control, many patients experience limited therapeutic efficacy and often suffer from adverse effects.^[2,3] Thus, understanding the pathophysiological mechanisms underlying AF, particularly at the molecular level, is essential for developing more effective treatment approaches.^[4–7]

Recent years have seen a growing interest in the role of cell death in the development of AF.^[8] Ferroptosis, a form of

iron-dependent cell death closely linked to inflammation and oxidative stress, has been implicated in the pathogenesis of various cardiovascular diseases.^[9,10] Although preliminary evidence suggests that ferroptosis may contribute to AF development, the specific mechanisms and key genes involved remain poorly understood.^[11,12]

This study aims to investigate the role of ferroptosis-related genes (FRGs) in AF using bioinformatics analysis. By analyzing datasets from the gene expression omnibus (GEO) database, we employ least absolute shrinkage and selection operator (LASSO) and support vector machine-recursive feature elimination (SVM-RFE) machine learning methods to identify key genes associated with AF. We further explore the potential functions and interaction networks of these genes. Additionally, we evaluate the diagnostic potential of these

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The datasets generated during and/or analyzed during the current study are publicly available.

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genes for AF using receiver operating characteristic curve (ROC) analysis and validate our findings in an independent AF dataset. This includes examining how these genes may influence AF development through interactions with miRNAs and lncRNAs, as well as their relationship with immune cell infiltration.

Through this comprehensive analysis, our study aims to elucidate the critical role of ferroptosis in AF pathogenesis and identify molecular targets for novel therapeutic strategies. Moreover, our research seeks to enhance the understanding of the interplay between cell death pathways in cardiovascular disease, providing a foundation for future research in this field (Fig. 1).

2. Materials and methods

2.1. Identification and correlation analysis of key ferroptosis-related genes

The term “AF” was used to search for gene expression profile datasets related to AF in the GEO database (<https://www.ncbi.nlm.nih.gov/geo/>). Datasets were included based on the

following criteria: whole-genome mRNA microarray data, samples sourced from heart tissue of AF and non-AF (control/healthy) subjects, limited to Homo sapiens, and (iv) sufficient sample size for analysis. Ultimately, 2 microarray datasets (GSE115574 and GSE79768) were downloaded from the GEO database, with probes annotated using the datasets’ respective annotation files. Differentially expressed genes (DEGs) were identified using R software (version 4.2.1) and the “limma” package, with significance defined as P -value $< .05$ and $|\log_2\text{FC}| > 0.5$. Ethical approval was not required as the GEO database contains publicly available data.

A total of 729 FRGs were downloaded from the FerrDb database (<http://www.zhounan.org/ferrdb/>), and the “limma” package was used to filter DEGs to identify FRGs. The “heatmap” and “corrplot” packages were employed to generate heatmaps and correlation plots, respectively.

2.2. Gene ontology and Kyoto Encyclopedia of Genes and Genomes pathway enrichment analyses of genes

Gene ontology (GO) and Kyoto Encyclopedia of Genes and Genomes (KEGG) functional enrichment analyses were conducted using the “clusterProfiler” package. Enrichment was based on the hypergeometric distribution, with analyses focusing on GO categories: molecular function (MF), cellular component (CC), and biological process (BP).

2.3. Screening of key ferroptosis-related genes with the least absolute shrinkage and selection operator and support vector machine-recursive feature elimination algorithms

The R package “glmnet” was used to construct the optimal LASSO logistic regression model. The best model was determined by selecting the tuning parameter (λ) that minimizes the mean squared error through cross-validation. LASSO is particularly effective for reducing model complexity and feature selection by shrinking the coefficients of less significant variables to zero, thereby identifying the most relevant predictors for AF.

The SVM-RFE algorithm was implemented using the R package “e1071.” Initially, the importance of the FRGs was ranked based on the feature weights of the support vectors. This ranking was further refined, and the cross-validation error was estimated using ten-fold cross-validation.

2.4. Analysis of receiver operating characteristic curves

Genes identified by both the LASSO and SVM-RFE models were intersected using a Venn diagram to identify common genes, referred to as interGenes. The “pROC” package in R software was used to conduct ROC curve analysis, assessing the stability of the 2 algorithms and their diagnostic sensitivity for AF.

2.5. Gene set enrichment analysis of interGenes

GSEA is commonly used to analyze and interpret coordinated pathway-level changes in transcriptomic experiments. In this study, GSEA was performed using the “clusterProfiler” package in R software, with visualization facilitated by the “ggplot2” and “enrichplot” packages. Gene sets were considered significant if they met the criterion of an adjusted P -value $< .05$.

2.6. Correlation between interGenes and infiltrating immune cells

The CIBERSORT method (<http://cibersort.stanford.edu/>) was employed to analyze mRNA expression profiles, characterizing

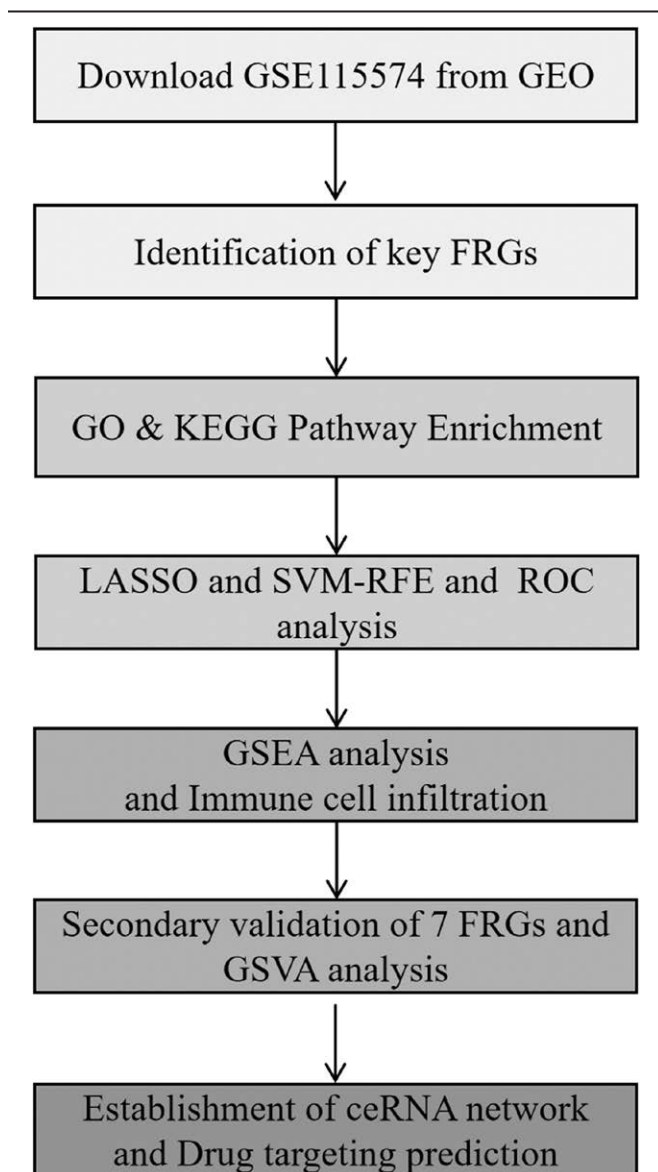


Figure 1. The overall protocol of this study.

the cellular composition of left atrial and normal tissues across 22 types of immune cells (including 7 types of T cells, naive and memory B cells, plasma cells, NK cells, and myeloid subsets). Samples with a CIBERSORT P -value $< .05$ were included in subsequent analyses. Visualization was conducted using the “reshape2,” “tidyverse,” “ggplot2,” and “vioplot” packages in R software, with significant differences indicated by $P < .05$.

2.7. Gene set variation analysis of interGenes

Gene set variation analysis is an unsupervised, nonparametric method for assessing transcriptome gene set enrichment. This study used the “c5.go. symbols” and “c2.cp.kegg. symbols” gene sets from molecular biology databases as reference sets. The “GSVA” package in R software was used to score GO functions and KEGG pathways in normal and AF samples, with $P < .05$ considered statistically significant.

2.8. Predict potential therapeutic drugs

To prioritize potential therapeutic drugs for AF, this study utilized the DGIdb database (<https://www.dgldb.org/>) to analyze interGenes as drug candidates. A drug-gene interaction network was constructed based on the prediction results using Cytoscape.

2.9. Validation of results

An additional AF dataset (GSE79768) was accessed from the GEO database and screened using the same criteria as GSE115574.

2.10. Gene-miRNA interactions and lncRNA prediction

mRNA-miRNA interactions were predicted using TargetScan (<http://www.targetscan.org/>), miRDB (<http://mirdb.org/>), and miRBase (<http://www.mirbase.org/>). Only miRNAs identified by all 3 prediction tools were considered crucial. The predicted miRNAs were then uploaded to StarBase 2.0 (<http://starbase.sysu.edu.cn/index.php>) with stringent filtering criteria (>5) to predict potential lncRNAs targeting these miRNAs. A lncRNA-miRNA-FRG regulatory network was constructed using Cytoscape software.

2.11. Statistical analysis

All statistical analyses were performed using R software (version 4.2.1, The R Foundation for Statistical Computing, Vienna, Austria) and RStudio (version 2024.1.1, Posit Software, Boston, MA, USA).

3. Result

3.1. Differential expression genes identification in atrial fibrillation

After merging and organizing the GSE115574 dataset, an expression matrix comprising 21,654 genes was obtained. Analysis of the FerrDb database identified a total of 73 FRGs (Table S1, Supplemental Digital Content, <http://links.lww.com/MD/N640>). Differential expression analysis of these FRGs revealed 37 upregulated FRGs and 36 downregulated FRGs (Fig. 2A).

3.2. Correlation between the expression of 73 ferroptosis-related genes in atrial fibrillation

Correlation analysis was performed on the 73 FRGs, leading to a correlation matrix relationship diagram. As shown

in Figure 2B, each square in the correlation matrix represents the correlation between the expressions of 2 FRGs. Red and blue indicate positive and negative correlations, respectively. Correlation coefficient is shown in the lower left area of the matrix, while its significance is shown in the upper right area.

3.3. Functions and pathways of ferroptosis-related genes

To further explore the primary MFs and signaling pathways of these 73 FRGs, this study conducted GO and KEGG enrichment analyses. GO enrichment analysis revealed that FRGs were enriched in different GO terms, including response to oxidative stress, cellular response to external stimulus, and mitochondrial outer membrane (Fig. 2C and D). KEGG enrichment analysis indicated that FRGs were primarily enriched in pathways of neurodegeneration, autophagy, MAPK signaling pathway, and ferroptosis (Fig. 2E).

3.4. 7 key ferroptosis-related genes were obtained by the 2 machine learning methods in total

Using the LASSO regression method, the following 24 FRGs were identified: GOT1, ALOX5, KEAP1, ATG7, ALOXE3, EGFR, MAPK14, ATF3, FBXW7, CTSB, BRD7, TSC1, PPARG, PAQR3, SLC7A11, ADAM23, HSPA5, JUN, CAV1, PARK7, ATF2, ALDH3A2, MTF1, CISD3 (Fig. 3A and B). Meanwhile, the SVM-RFE algorithm identified the following 8 FRGs: CISD3, ALOX5, PARK7, MAPK14, CAV1, VDAC2, ADAM23, and TSC1 (Fig. 3C and D). Based on the FRGs obtained from both algorithms, the intersection yielded 7 key FRGs: ALOX5, MAPK14, TSC1, ADAM23, CAV1, PARK7, and CISD3 (Fig. 3E).

3.5. Receiver operating characteristic curves of the 7 ferroptosis-related genes and their logistic regression model

Based on the 7 key FRGs obtained from the 2 machine learning methods, a logistic regression model was first established, followed by plotting the model's ROC curve. The AUC values for ALOX5, MAPK14, TSC1, ADAM23, CAV1, PARK7, and CISD3 were 0.675, 0.764, 0.658, 0.699, 0.696, 0.653, and 0.695 respectively (Fig. 4A). The diagnostic model's AUC value was 0.957, indicating a high diagnostic accuracy and reliability of the genes (Fig. 4B).

3.6. Signaling pathways identified by gene set enrichment analysis

gene set enrichment analysis was performed based on the data of the 7 key FRGs, as shown in Figure 4C. Among the activated pathways, pathways such as ECM receptor interaction and TGF-beta signaling pathway were primarily enriched in the AF group.

3.7. Differential immune cell infiltration in atrial fibrillation

Differences in immune cell infiltration were observed between the control group and the AF group (Table S2, Supplemental Digital Content, <http://links.lww.com/MD/N640>). Compared to the control group, the number of B cells memory was significantly increased in the AF group ($P < .05$). Similarly, the quantities of T cells regulatory and Monocytes were significantly reduced in the AF group compared to the control group ($P < .05$) (Fig. 5A).

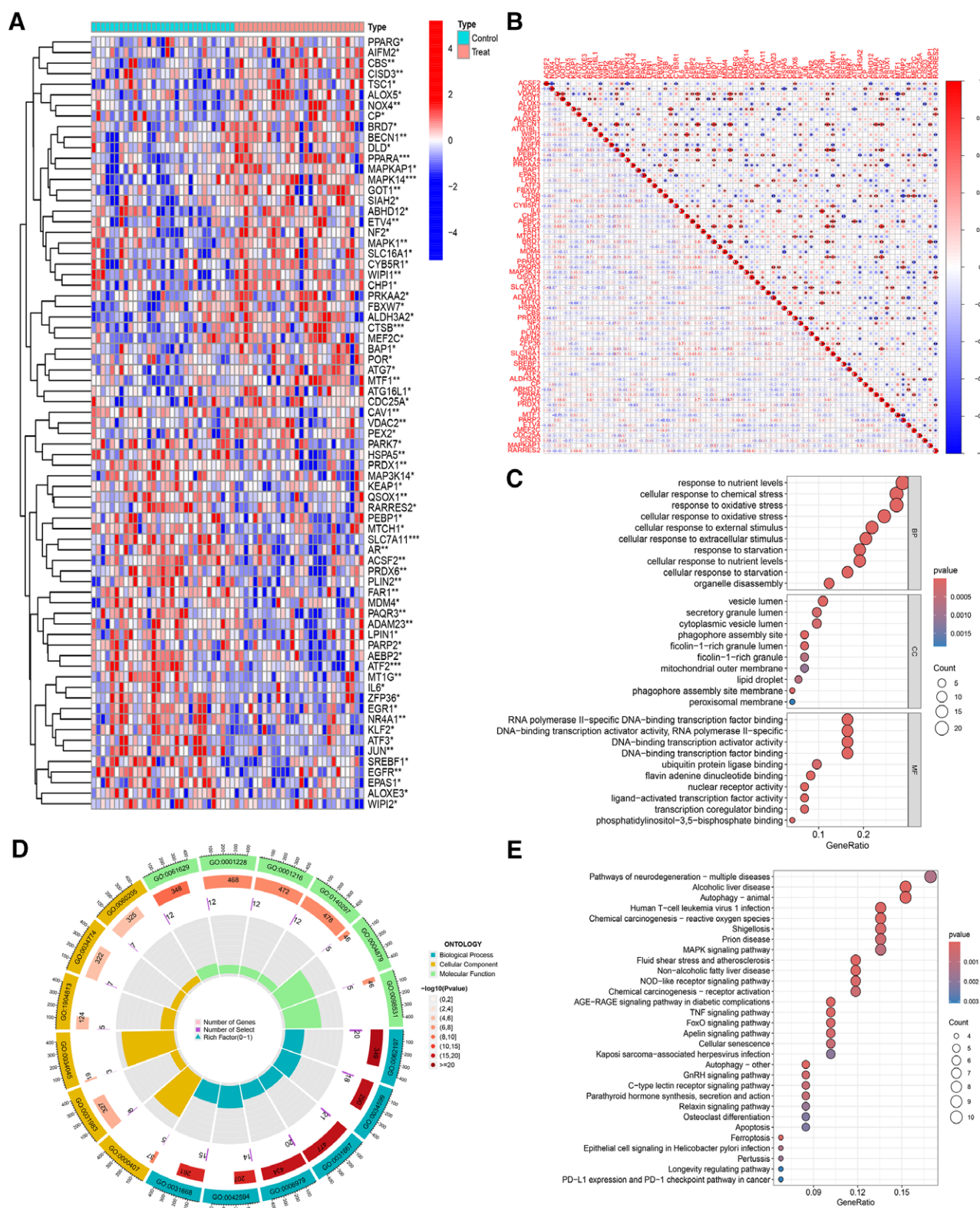


Figure 2. The gene differential expression in atrial fibrillation. (A) The heatmap of 73 FRGs in GSE115574. (B) Correlation between the expression of 73 FRGs. (C-E) The main GO terms and KEGG pathways enriched by differentially expressed mRNAs.

3.8. Correlations between the expression of the 7 ferroptosis-related genes and the infiltration levels of 22 types of immune cells in atrial fibrillation

A correlation heatmap was generated based on the expression levels of the 7 key FRGs and the infiltration levels of 22 types of immune cells (Fig. 5B). The expression level of ADAM23 was

positively correlated with Dendritic cells activated and NK cells activated, and negatively correlated with Macrophages M1 and NK cells resting. The expression level of ALOX5 was positively correlated with T cells CD8. The expression level of CAV1 was positively correlated with Dendritic cells activated and negatively correlated with Dendritic cells resting. The expression level of MAPK14 was positively correlated with Eosinophils

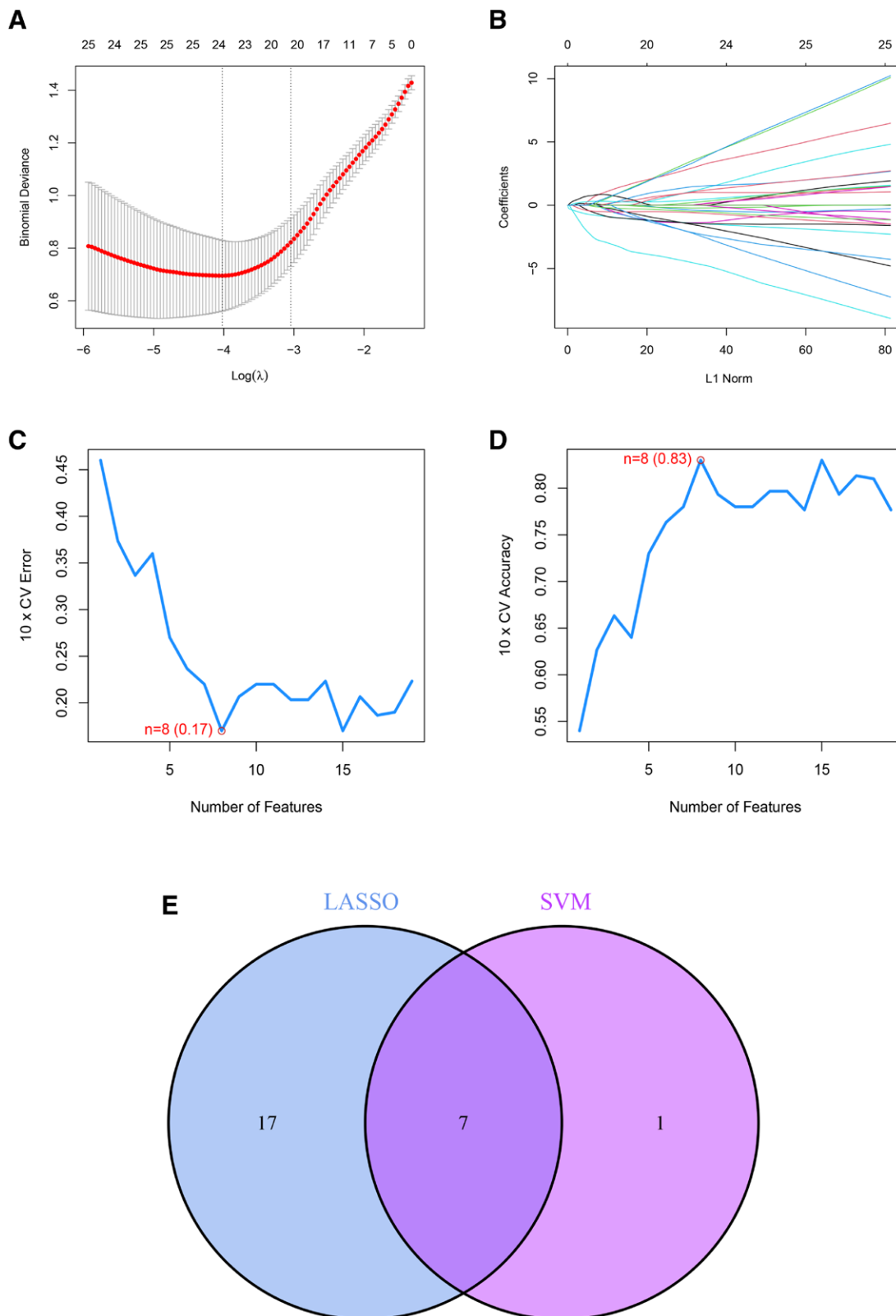


Figure 3. Getting key FRGs through machine learning. (A and B) The performance in of ten-time cross-verification for tuning parameter in selection least absolute shrinkage and selection operator (LASSO). Each coefficient curve in the upper picture represents a single gene. The solid vertical lines in another picture represent the partial likelihood deviance SE, and the number of genes ($n = 10$) corresponding to the lowest point of the curve is the most suitable for LASSO. (C and D) Accuracy graph and cross-validation error graph of FRGs screened with the SVM-RFE algorithm. (E) Venn diagram of the intersection of FRGs screened with the 2 machine learning methods.

and negatively correlated with Plasma cells and T cells regulatory. The expression level of PARK7 was positively correlated with Macrophages M1 and NK cells resting, and negatively

correlated with NK cells activated. The expression level of TSC1 was positively correlated with NK cells activated and T cells follicular helper (P all $< .05$).

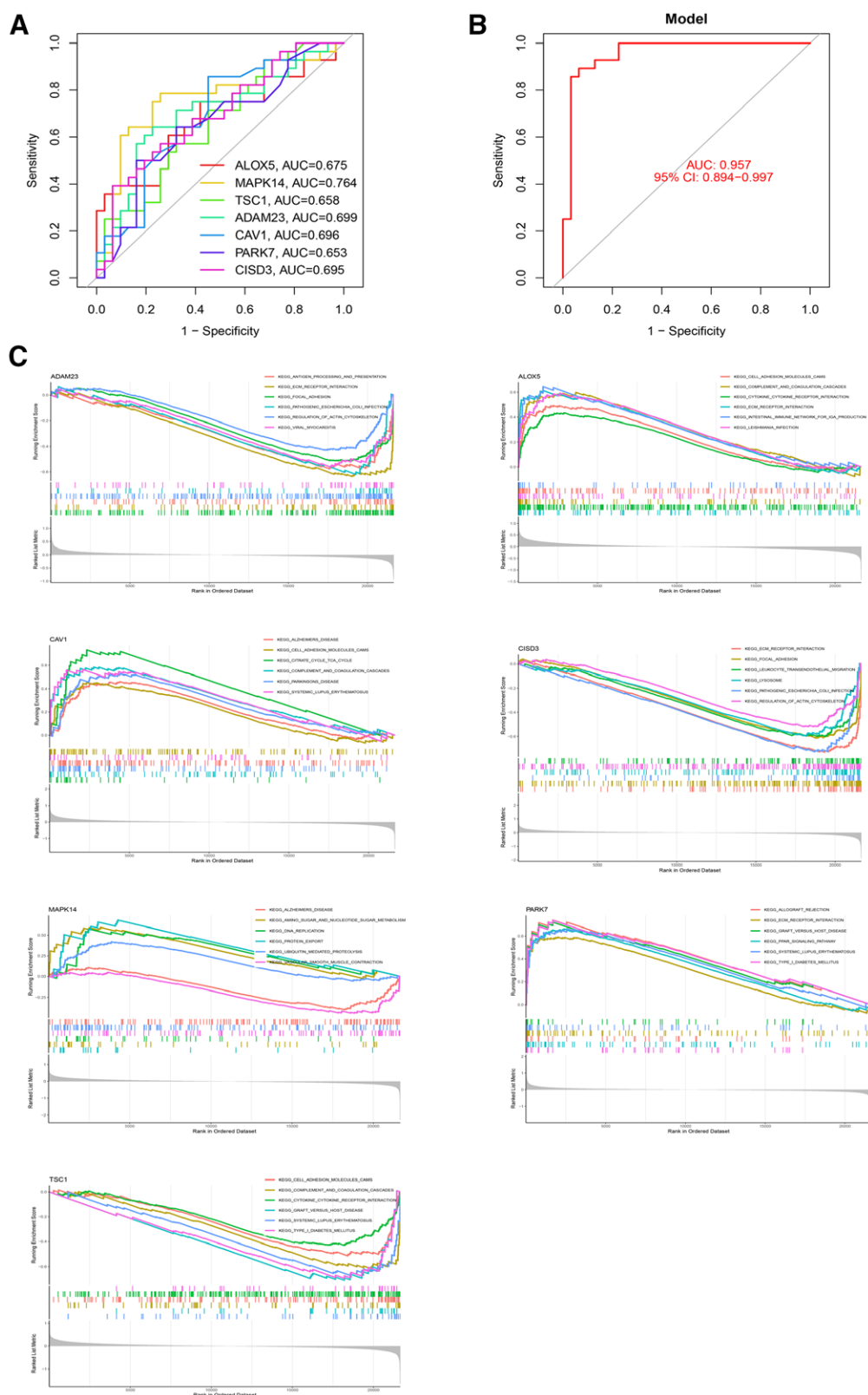


Figure 4. ROC curves and GSEA results for groups. (A) ROC curves of the 7 FRGs. (B) Logistic regression model (AUC value was 0.957). (C) Plots of GSEA results for groups with high expression of FRGs.

3.9. Metabolic pathway involvement and correlation with shared hub gene

The GSVA results of the metabolic pathways related to the 7 FRGs are displayed in a heatmap (Fig. 6A). The TGF-beta

signaling pathway and ECM receptor interaction were the most prominent among the upregulated signaling pathways, aligning with the results obtained from previous comprehensive analyses. The pathway of primary bile acid biosynthesis was the only downregulated signaling pathway.

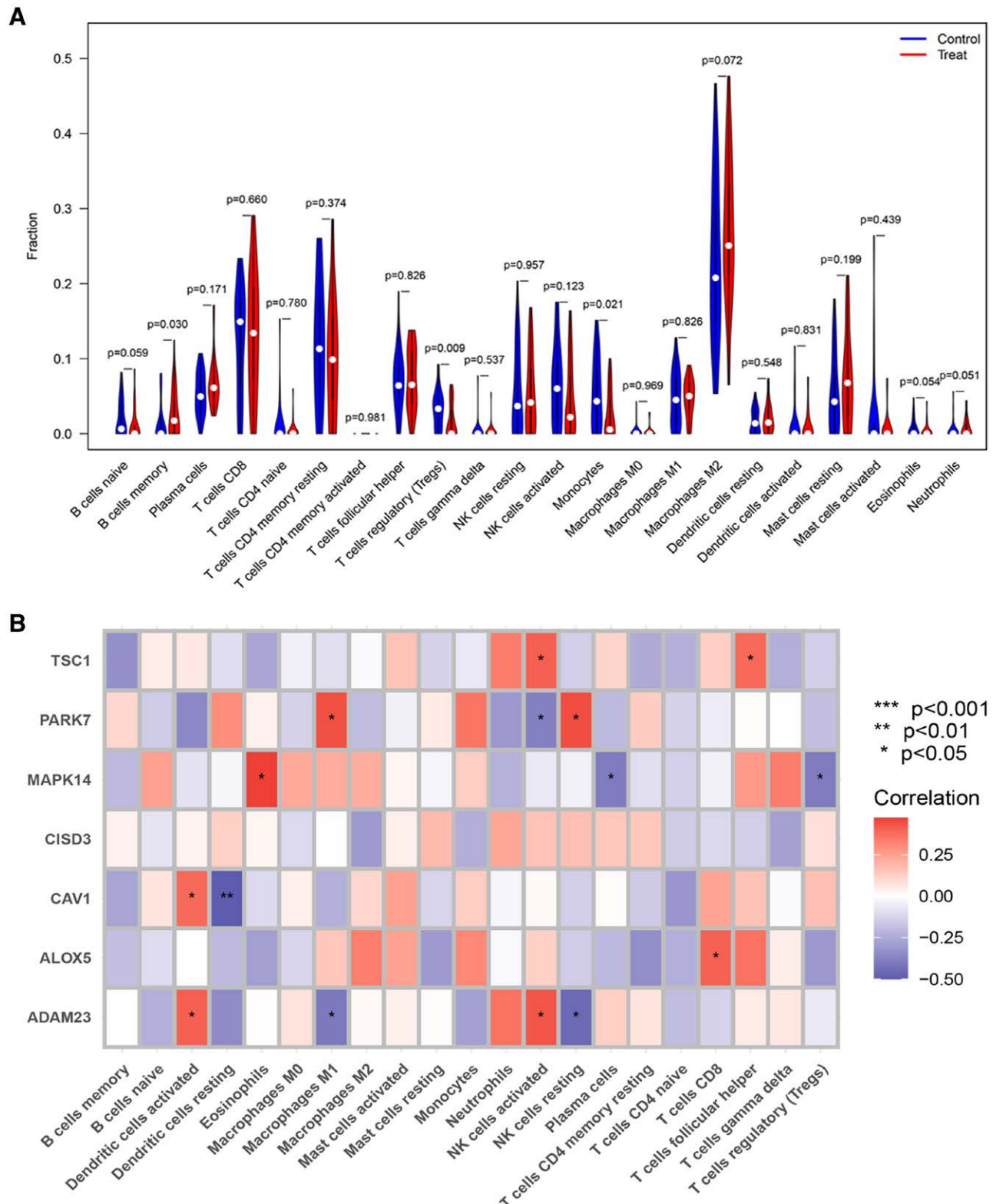
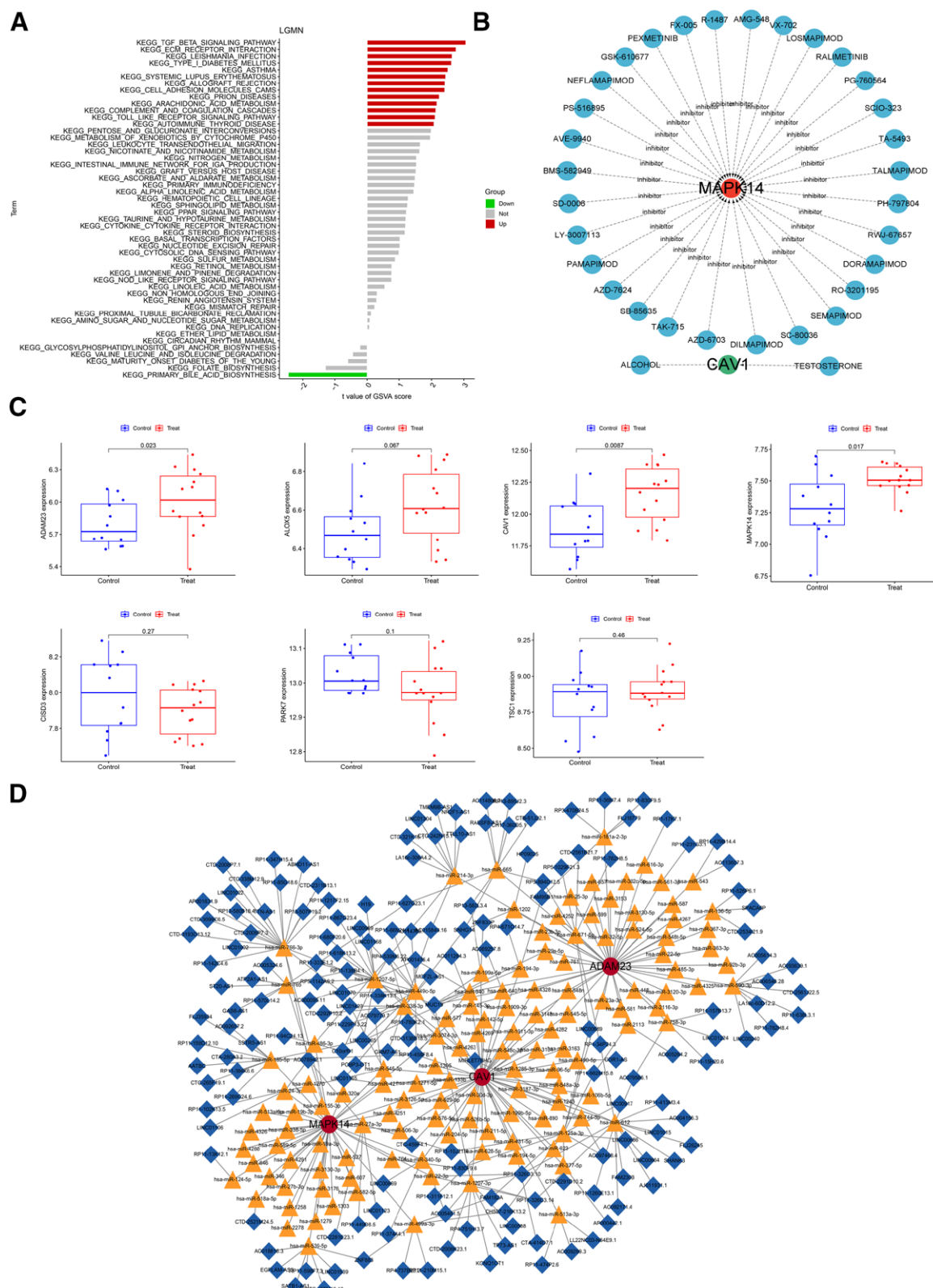


Figure 5. Differential immune cell infiltration in AF. (A) Differences in the infiltration of 22 types of immune cells in AF. Treat is AF. (B) Heat map of the correlation between the expression levels of the 7 FRGs and the infiltrating numbers of 22 types of immune cells in each group. Red squares: positive correlation; purple squares: negative correlation. Darker color indicates stronger correlation. Not significant (ns), * $P < .05$, ** $P < .001$, *** $P < .001$.

3.10. Targeted drug prediction

This study utilized the Dgibid database to predict potential target drugs related to key genes, which may aid in treating AF by modulating ferroptosis. Ultimately, 102 target drugs

were predicted. Among these, 25 potential drugs are related to ALOX5, 72 to MAPK14, and 3 to TSC1, with another 3 related to TSC1, and 2 potential drugs related to CAV1 (Fig. 6B).



220 potential lncRNAs targeting these miRNAs were identified, leading to the construction of a ceRNA network diagram (Fig. 6D).

4. Discussion

This study delves into the latest advancements in the understanding of ferroptosis and AF, marking a significant step forward by being the first to integrate LASSO regression and SVM-RFE machine learning techniques to comprehensively explore the bioinformatic landscape of FRGs in AF. In contrast to prior AF research, this study not only employs functional enrichment analysis, immune cell infiltration analysis, and ceRNA network construction to thoroughly elucidate the molecular mechanisms underpinning AF but also validates the miRNA and lncRNA regulatory networks of key genes using independent datasets and predicts potential drug targets. This approach significantly enhances the scientific rigor and clinical relevance of the study's findings.

The implementation of LASSO and SVM-RFE machine learning algorithms allowed for the refinement of the FRG list, confirming their predictive capacity for AF. The genes identified by both algorithms, particularly the 7 key FRGs, emerged as robust biomarkers with potential diagnostic value for AF. These FRGs were further validated through ROC curve analyses, underscoring the reliability of these genes as predictive markers for AF and demonstrating the efficacy of the rigorous bioinformatics screening employed in this study.

The TGF-beta signaling pathway has been identified as a critical factor in the pathogenesis of AF, with the results of the GSEA analysis in this study reinforcing this connection.^[13–15] Dysregulation of the TGF-beta pathway has been implicated in AF-related structural changes in the heart, such as fibrosis, highlighting its significance in disease progression and as a potential therapeutic target.

Immune cells, as essential components of the immune system, may also play a pivotal role in the immunological mechanisms underlying AF. This study's analysis of immune cell infiltration in DEGs revealed alterations in immune cell populations, including an increase in memory B cells and a decrease in regulatory T cells and monocytes in the atrial tissues of AF patients compared to controls. These findings align with previous studies reporting abnormalities in CD33 + myeloid cells, CD14 + monocytes, activated dendritic cells, and regulatory T cells (Tregs) in the atrial tissues of AF patients.^[16,17] These immune cell changes provide a theoretical foundation for future exploration of the immunological processes involved in AF.^[18,19]

Through repeated validation of the FRGs, the expression levels of ADAM23, CAV1, and MAPK14 were consistently found to be closely associated with AF. CAV1 plays a critical role in regulating multiple signaling pathways, including interactions with G protein-coupled receptors, tyrosine kinases, and endothelial nitric oxide synthase.^[20–22] Genetic variants linked to an increased risk of AF have been shown to potentially regulate CAV1 expression.^[23] CAV1 may also promote atrial fibrosis through the STAT3 signaling pathway, thereby increasing susceptibility to AF.^[24,25] MAPK14 is a key component of the MAPK signaling pathway and is involved in stress response, cell differentiation, and apoptosis.^[26] Meanwhile, ADAM23 functions as a protease and is implicated in cell-to-cell interactions and signaling. It plays a role in the degradation and remodeling of the extracellular matrix, which is crucial in the structural alterations of the heart, particularly in the atria, that contribute to the development of AF.^[27,28] By regulating the composition of the extracellular matrix, ADAM23 may influence the mechanical properties of cardiac tissues, indirectly affecting the onset and progression of AF. Additionally, ADAM23 may be involved in the pathogenesis of AF by modulating inflammatory and

immune responses, given the role of ADAM family proteins in these processes.^[29,30]

The prediction of miRNAs and lncRNAs that target and regulate ADAM23, CAV1, and MAPK14 was conducted using bioinformatics tools, leading to the construction of a ceRNA network. This network not only enhances our understanding of the gene interactions and regulatory mechanisms in AF but also identifies potential therapeutic targets.^[31] By targeting specific miRNAs or lncRNAs, it may be possible to modulate the expression of ADAM23, CAV1, and MAPK14, offering new avenues for AF treatment. Additionally, studying the ceRNA network could shed light on the molecular mechanisms involved in ferroptosis, providing a theoretical foundation for the development of novel drugs targeting ferroptosis-related diseases.

This study also utilized drug databases to predict potential drugs targeting AF. While most of these drugs are primarily associated with cancer, asthma, and inflammatory diseases and have limited direct relevance to AF treatment, emerging research may uncover their potential role in AF therapy. Among these, drugs such as Montelukast, Rilopirox, and Tenidap exhibit anti-inflammatory properties. Montelukast is already approved for treating asthma, and although the others have not yet received clinical approval, their anti-inflammatory effects could make them potential candidates for AF treatment. Additionally, drugs like Methotrexate, an immunosuppressant mainly used for conditions such as rheumatoid arthritis, also possess anti-inflammatory properties, suggesting they may hold promise in AF therapy.

Despite these promising findings, the study recognizes its limitations. The datasets analyzed were retrospective and publicly sourced, which could introduce selection bias. Moreover, while the bioinformatic approach was robust, validation through *in vitro* studies and clinical trials is imperative for translating these findings into actionable therapeutic strategies.

5. Conclusion

In this study, we identified key FRGs that may serve as therapeutic targets for AF through comprehensive bioinformatics analysis. By integrating multiple datasets and employing advanced machine learning techniques such as SVM-RFE and LASSO, we were able to pinpoint specific genes that play a pivotal role in the pathogenesis of AF. Our findings provide new insights into the molecular mechanisms underlying AF and suggest potential therapeutic strategies targeting ferroptosis pathways. Future experimental validation and clinical studies are necessary to explore the feasibility of these targets in AF treatment. This research not only contributes to the growing body of knowledge on AF but also opens new avenues for the development of targeted therapies that could improve patient outcomes.

Author contributions

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Writing – review & editing: Shaoyi Peng, Kaiyuan Li, Yilong Man, Peng Liu.

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